Separability Splits Length from Geodesic Structure in Nonlinear Lebesgue Spaces

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11 August 2026

Status: candidate partial result, likely valid. For the length-space conclusion in Proposition 4.4 of arXiv:2603.09459, global separability of the complete target is unnecessary. In contrast, separability cannot be removed from the geodesic conclusion in general: for every $p \in (1, \infty]$ there is a complete nonseparable geodesic target N for which the corresponding nonlinear L^p space is not geodesic. The possibility of dropping completeness remains open.

1 Source and question

Guillaume Sérieys, *Nonlinear Lebesgue spaces: Curves and geometry*, arXiv:2603.09459 (2026), defines $L^p_h(M, N)$ using equivalence classes of measurable maps with separable range and the metric

$$D_p(f, g) = \left(\int_M d_N(f(x), g(x))^p \, d\mu(x) \right)^{1/p}$$

for finite p, with the usual essential-supremum definition for $p = \infty$. Proposition 4.4 proves that this space is a length (respectively geodesic) space when N is complete, separable, and length (respectively geodesic).

On arXiv PDF page 31 the source records the remaining issue [1]:

Source question (Remarks 4.5–4.6). *Can the completeness and separability assumptions on N in Proposition 4.4 be relaxed?*

When N is a geodesic space, the proof works with $k = 1$. □

Remark 4.5 (On the separability assumption on N)**.** Note that the proof of Proposition 4.4 only requires the completeness and separability assumptions on N to ensure that the pointwise selection of curves is measurable. When N is uniquely geodesic and d_N is jointly convex, for instance, when N is a global NPC space (Definition 4.10), N need not be separable for the conclusion of Proposition 4.4 to hold since there is no measurable selection of geodesics to construct as any pair of points in N is joined by a unique geodesic and geodesics in N are continuous with respect to their endpoints thanks to the joint convexity of d_N. The question of whether the assumptions of completeness and separability on N can be relaxed remains open.

Remark 4.6 (Related results in the literature)**.** We emphasize that while the conclusion of Proposition 4.4 in the geodesic case already appeared in a work (under the assumption that h is constant) by J. Jost [Jos97b, Lemma 4.1.2], the proof seemed incomplete as it selected pointwise geodesics without checking the measurability with respect to endpoints, which is precisely what is done in the proof of Proposition 4.4, hence missing the key assumptions of completeness and separability on N. As highlighted in Remark 4.5, the question of whether the completeness and separability assumptions on N can be relaxed remains open.

The proof in the source uses completeness and separability to select a measurable family of nearly minimizing pointwise curves. The following results show that the two conclusions behave differently.

2 Results

Theorem 1 (Global separability is unnecessary for length spaces). *Let (M, Σ, μ) be any measure space, let $h \in L^0(M, N)$, and let $p \in [1, \infty]$. If N is a complete length space, with no separability assumption, then $L^p_h(M, N)$ is a length space.*

Theorem 2 (Separability cannot generally be dropped for geodesicity). *For every $p \in (1, \infty]$ there exist a probability space (M, Σ, μ), a complete nonseparable geodesic metric space N, and $h \in L^0(M, N)$ such that $L^p_h(M, N)$ is not a geodesic space. In the same examples it is a complete length space by Theorem 1.*

Thus separability can be removed entirely from the length half of the source proposition, but not from its geodesic half for $p > 1$. These statements do not decide whether completeness can be dropped.

3 Proof intuition

The positive result uses the strong measurability built into the definition: the base map and any two endpoints already have separable ranges. A countable approximate-midpoint closure of those ranges sits inside a closed, separable, complete length subspace of N. The source's measurable-selection argument can then be run in that smaller target.

For the negative result, use an uncountable ladder. The two rails are separable, so the endpoint maps are admissible; the midpoint of each vertical rung lies in an uncountable uniformly discrete set. Strict convexity of the scalar L^p norm for $p > 1$ forces any function-space midpoint to choose those pointwise rung midpoints almost everywhere, making its range nonseparable.

4 A separable length hull

Call z an ε-midpoint of x, y if

$$\max\{d(x, z), d(z, y)\} \leq \tfrac{1}{2} d(x, y) + \varepsilon.$$

Lemma 3 (Separable reduction). *If X is a complete length space and $A \subset X$ is separable, then there is a closed separable length subspace $Y \subset X$ containing A.*

Proof. Choose a countable set S_0 dense in A. Given a countable S_n, for every $u, v \in S_n$ and every $k \geq 1$, choose a $1/k$-midpoint $m(u, v, k)$ in X, and put

$$S_{n+1} = S_n \cup \{m(u, v, k) : u, v \in S_n,\ k \geq 1\}.$$

Such midpoints exist because a curve from u to v whose length is arbitrarily close to $d(u, v)$ has a point arbitrarily close to half its length. Let

$$Y = \overline{\bigcup_{n \geq 0} S_n}^{,X}.$$

Then Y is closed and separable, contains A, and is complete because X is complete.

It remains to show that Y is a length space. Given $y, z \in Y$ and $\varepsilon > 0$, choose u, v in the countable union at a common sufficiently late stage with $d(y, u), d(z, v) < \delta$. At the next stage choose $m(u, v, k)$ with $1/k < \delta$. The triangle inequality gives

$$\max\{d(y, m), d(m, z)\} \leq \tfrac{1}{2} d(y, z) + 3\delta.$$

As δ is arbitrary, Y has approximate midpoints with every positive error. The standard dyadic construction now gives, for every $\eta > 0$, a curve in Y from y to z of length at most $d(y,z) + \eta$: recursively insert approximate midpoints with summable errors, define the curve first on dyadic times, and use completeness to extend it continuously. Hence Y is a length space. □

Proof of Theorem 1. Choose separably valued representatives of h, f, g, where f, g are arbitrary elements of $L^p_h(M,N)$. Their three ranges have separable union A. Apply Lemma 3 to obtain a closed, separable, complete length subspace $Y \subset N$ containing those ranges.

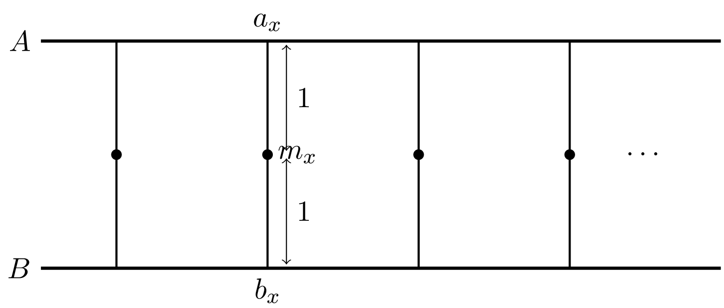
Now regard h, f, g as Y-valued maps. Proposition 4.4 of the source, applied to Y, gives for every $\kappa > 1$ an absolutely continuous curve c from f to g in $L^p_h(M,Y)$ with

$$L(c) \leq \kappa D_p(f,g).$$

The inclusion $Y \hookrightarrow N$ induces an isometric inclusion of the corresponding nonlinear Lebesgue spaces, so c is also an admissible curve in $L^p_h(M,N)$. Letting $\kappa \downarrow 1$ proves that its intrinsic metric equals D_p. □

5 The uncountable ladder counterexample

Let $A = \{a_x : x \in [0,1]\}$ and $B = \{b_x : x \in [0,1]\}$ be two copies of the unit interval, each with its usual length metric. For every $x \in [0,1]$, attach a distinct interval E_x of length 2, joining a_x to b_x. Give the resulting metric graph N its intrinsic path metric.



Lemma 4 (Geometry of the ladder)**.** *The space N is complete and geodesic. The edge E_x is the unique geodesic from a_x to b_x. If m_x is its midpoint, then $d_N(m_x, m_y) \geq 2$ whenever $x \neq y$.*

Proof. A shortest path can be reduced to one which uses at most one complete vertical edge: any two unnecessary rail changes cost 4 and can be replaced by travel along a rail of length at most 1. Among the remaining candidates, the possible crossing point belongs to the compact interval $[0,1]$ and the path length depends continuously on it, so a minimum is attained. Thus N is geodesic.

For a_x and b_x, the direct edge has length 2. A path crossing at $z \neq x$ has length at least $|x - z| + 2 + |z - x| > 2$, proving uniqueness. A path from m_x to m_y with $x \neq y$ must first travel distance 1 to an endpoint of E_x and finally distance 1 from an endpoint of E_y, so the midpoints are 2-separated.

Finally, let (q_n) be Cauchy. If it stays a fixed positive distance from both rails, it must eventually lie in one vertical edge, because interiors of distinct such edges are uniformly separated there. Otherwise it approaches one rail (it cannot alternate between the two, which are distance 2 apart), and its nearest rail coordinates form a Cauchy sequence in $[0,1]$. In both cases it converges in N, proving completeness. □

Proof of Theorem 2. Take $M = [0, 1]$ with Lebesgue measure and define

$$h(x) = f(x) = a_x, \qquad g(x) = b_x.$$

The ranges of f and g are the two separable rails, so these maps are admissible, and $D_p(f, g) = 2$ for every p.

Assume first that $1 < p < \infty$ and that a constant-speed geodesic exists. Let w be its value at time $1/2$, represented by a separably valued map, and write

$$r(x) = d_N(a_x, w(x)), \qquad s(x) = d_N(w(x), b_x).$$

Then $r + s \geq 2$ pointwise and $\|r\|_p = \|s\|_p = 1$. Since the base measure has mass one,

$$2 = \|2\|_p \leq \|r + s\|_p \leq \|r\|_p + \|s\|_p = 2.$$

Equality in Minkowski for nonnegative functions in L^p, $p > 1$, forces $r = s$ almost everywhere; their norms then give $r = s = 1$ almost everywhere. Together with $r + s = d(a_x, b_x)$, this says that $w(x)$ is the midpoint of a geodesic from a_x to b_x. Lemma 4 makes it the unique point m_x.

For $p = \infty$, midpoint equality gives $r, s \leq 1$ almost everywhere, while $r + s \geq 2$, so again $r = s = 1$ and $w(x) = m_x$ almost everywhere.

Every full-measure subset of $[0, 1]$ is uncountable, whereas every 2-separated subset of a separable metric space is countable. Hence the essential range $\{m_x\}$ of w is not separable, contradicting the defining requirement for membership in $L^p_h(M, N)$. No geodesic exists. □

Remark 5 (Why $p = 1$ is different)**.** *The counterexample is sharp with respect to the strict-convexity step. For $p = 1$, define $c(t)(x) = b_x$ for $x \leq t$ and $c(t)(x) = a_x$ for $x > t$. Then $D_1(c(s), c(t)) = 2|s - t|$, so c is a geodesic whose value at every time has separable range. This agrees with the source's observation that L^1-geodesics need not be pointwise geodesics.*

6 Verification, limitations, and novelty

The separable-hull proof was checked at the three points on which it depends: the construction is countable; closure preserves completeness inside the complete target; and the approximate-midpoint property passes from the countable core to the closure with explicit 3δ error. The ladder was checked directly for completeness, existence of minimizing paths, uniqueness of the designated rungs, and uniform separation of their midpoints. The L^p obstruction uses only equality in the scalar Minkowski inequality (or the elementary essential-supremum inequality at $p = \infty$).

This is intentionally a partial answer. It removes separability from the length conclusion and disproves its general removal from the geodesic conclusion for $p > 1$, but it does not decide whether completeness can be relaxed. It also makes no negative claim for the geodesic conclusion at $p = 1$.

A bounded novelty search through 11 August 2026 checked the run registry, exact source wording and title, approximate-midpoint separable reductions, separable length subspaces, metric-valued L^p geodesicity, nonseparable targets, and measurable geodesic selection. It found the source and general selection-theorem literature but no matching theorem or ladder counterexample. Novelty confidence is moderate and expert review is recommended, with special attention to Lemma 3 and the metric-graph completeness argument.

References

[1] G. Sérieys, *Nonlinear Lebesgue spaces: Curves and geometry*, arXiv:2603.09459, 2026.